

Report_DEB

1. Importation des données

```
file_path <- here::here("mod/FINAL/D_DEB_AWKe")

l_Experiments <- c(
  "Bart2019_XP1_1",
  "Bart2019_XP1_2",
  #"Bart2019_XP2",
  "Bart2019_XP3",
  "Bart2019_XP4_1",
  "Bart2019_XP4_2",
  "Bart2019_XP5",
  "Bart2020",
  #"Gollot2026_lufa"
  "Gollot2026_dens_D1",
  "Gollot2026_dens_D2",
  "Gollot2026_dens_D2b",
  "Gollot2026_dens_D5",
  "Gollot2026_dens_D7"
)

Adults_alone <- FALSE

df_data <- f_import_data_DEB(l_Experiments, Adults_alone)
```

2. Model fit

```

NbIter <- 100000
#seeds <- c(C1 = 3336, C2 = 6663, C3 = 1222, C4 = 2420, C5 = 4848, C6 = 5656, C7 = 7077, C8 = 8484)
seeds <- c(C1 = 3313, C2 = 6636, C3 = 2322)

text_priors <- '
Distrib (Fm,          TruncNormal_cv,          0.1,          0.2,          0.0001,          0.8);
Distrib (pAm,          TruncNormal_cv,          100,          0.2,          40,          2000);
Distrib (r_pAm_pM,      TruncNormal_cv,          0.8,          0.1,          0.1,          2);
Distrib (v,             TruncNormal_cv,          0.18505,        0.1,          0.001,          2);
Distrib (kapJ,          Uniform,                  0,              1);
Distrib (kapA,          Uniform,                  0,              1);
Distrib (Ehp,           TruncNormal_cv,          100,          0.2,          40,          300);
Distrib (E_coc,         TruncNormal_cv,          30,          0.2,          1,          300);
Distrib (rOM_ClxHorse,  LogUniform,          0.00001,        0.3);
Distrib (K,             Uniform,                  0.00001,        2);
Distrib (dv,           TruncNormal_cv,          2,            0.5,          0,          10);
Distrib (de,           LogUniform,          0.0000000001,    0.1);
Distrib (tau,          TruncNormal_cv,          30,          0.1,          0,          100);

# Calcul des vraisemblances
Distrib (Sigma_W, TruncNormal, 5, 1, 0.01, 100);'

text_likelihood <- 'Likelihood (Weight,          Normal, Prediction(Weight), Sigma_W);
Likelihood (Reproduction, Poisson, Prediction(Reproduction));'

OM_diff <- TRUE
RTOL <- 1e-7
ATOL <- 1e-6

# makemcsim DEBcali.model

f_In_tot(file_path, l_Experiments, Adults_alone, OM_diff, text_priors, text_likelihood, NbIter, RTOL, ATOL)

# ./mcsim.DEBcali DEB_C1.in

```

OM_horse and OM_soil distinct ? TRUE

Adult and alone earthworm taken into account ? FALSE

3. Results

	Fm.1.	pAm.1.	r_pAm_pM.1.	v.1.	kapJ.1.	kapA.1.
Result_mode	0.10258000	142.19800	0.77850000	0.18607300	0.91210000	0.5063740
2.5%	0.06508190	133.34205	0.65325900	0.15481700	0.87786410	0.4289020
97.5%	0.13446700	180.67400	0.94766100	0.22386095	0.94316800	0.6192300
Min	0.03130030	84.30800	0.58482400	0.11106200	0.79752900	0.0728887
Max	0.17272300	210.54600	1.02292000	0.25927700	0.96371300	0.6730070
Result_mean	0.09995576	154.85307	0.80958526	0.18941733	0.91413878	0.5256904
Result_sd	0.01752881	12.10688	0.07380172	0.01776677	0.01684099	0.0489991

	Ehp.1.	E_coc.1.	rOM_ClxHorse.1.	K.1.	dv.1.	de.1.
Result_mode	100.44100	25.37910	0.010105100	1.1394000	3.888460	1.434320e-09
2.5%	71.28210	21.75499	0.003041920	0.1605080	2.993930	1.434320e-09
97.5%	142.24080	35.72035	0.014834555	1.9166480	4.851903	9.858180e-06
Min	49.42320	17.46900	0.000373047	0.0638658	0.686175	1.084880e-09
Max	175.67900	41.15270	0.157797000	1.9997100	5.640320	1.239280e-05
Result_mean	105.40666	28.16131	0.008820690	0.8959061	3.789300	1.816045e-06
Result_sd	18.13578	3.53954	0.003408879	0.5174089	0.468336	2.957874e-06

	tau.1.	Sigma_W.1.
Result_mode	34.308900	0.1442260
2.5%	28.306400	0.1276583
97.5%	39.089200	0.1611819
Min	23.140100	0.1164160
Max	43.250700	5.1652500
Result_mean	33.781100	0.1436277
Result_sd	2.724292	0.0434561

Number of observations, n = 277

Number of parameters, k = 13

Log-likelihood, LL = -46.91721

BIC = 166.9466

Potential scale reduction factors:

	Point est.	Upper C.I.
de.1.	1.02	1.03
dv.1.	1.16	1.48
E_coc.1.	1.13	1.40
Ehp.1.	1.03	1.10
Fm.1.	1.01	1.02
K.1.	1.01	1.02

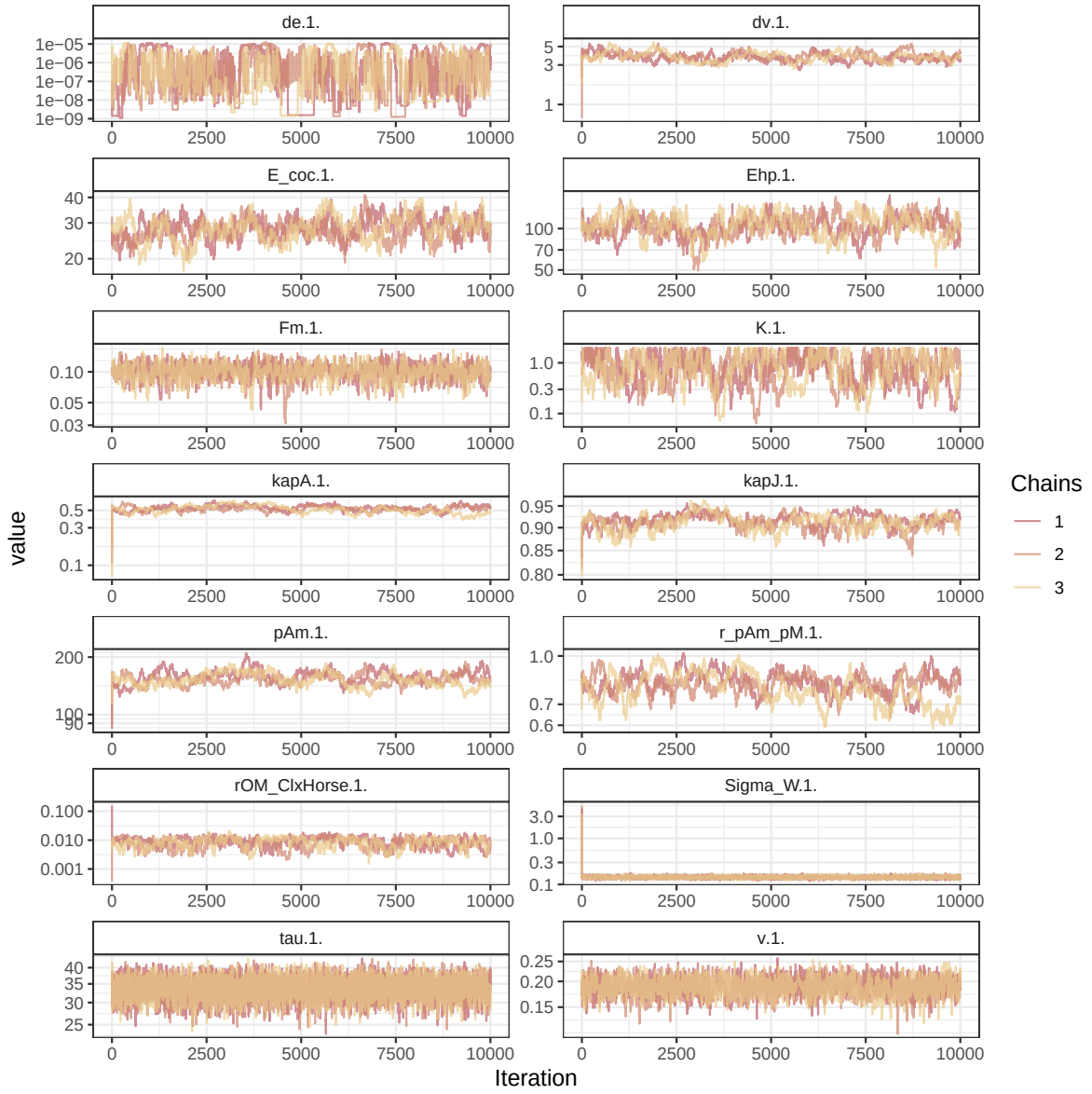


Figure 1: Parameters chains. Only the first iteration and the last `Nb_iter_kept` are represented.

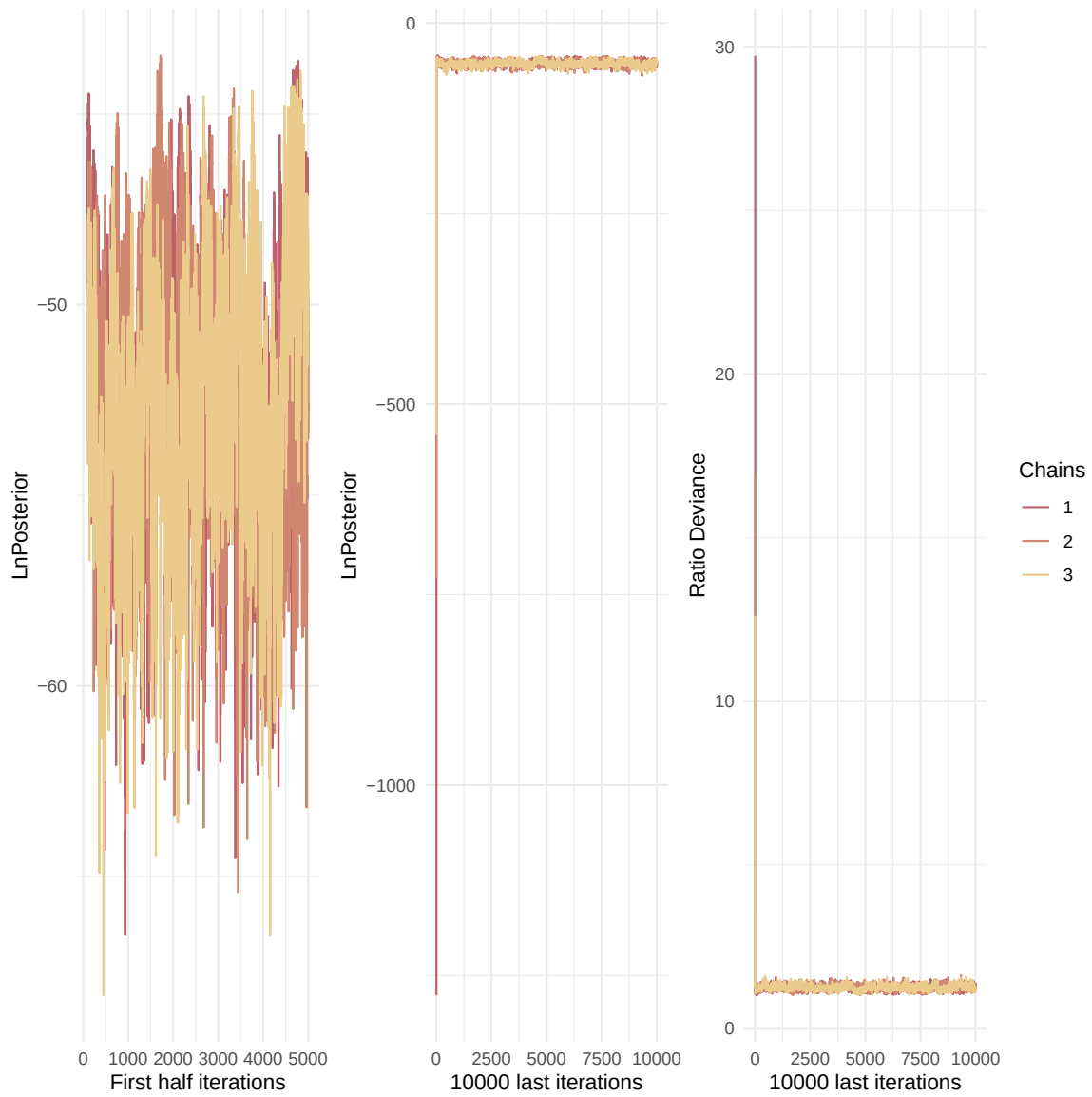


Figure 2: Evolution of likelihood and deviance ratio through the iterations.

kapA.1.	1.50	2.27
kapJ.1.	1.13	1.39
pAm.1.	1.23	1.63
r_pAm_pM.1.	1.53	2.33
rOM_ClxHorse.1.	1.02	1.05
Sigma_W.1.	1.01	1.02
tau.1.	1.00	1.00
v.1.	1.01	1.05

Multivariate psrf

1.36

```
mcmc_list_corr <- mcmc.list(
  lapply(mcmc_list, function(x) {
    mcmc(as.matrix(x)[-1, , drop = FALSE]) # <- reconvertir en mcmc
  })
)

colramp_aurora <- colorRampPalette(rev(Nord_aurora[1:4]))
ipairs(as.matrix(mcmc_list_corr), colramp = colramp_aurora, pixs=0.25, cex.diag = 0.5)
```

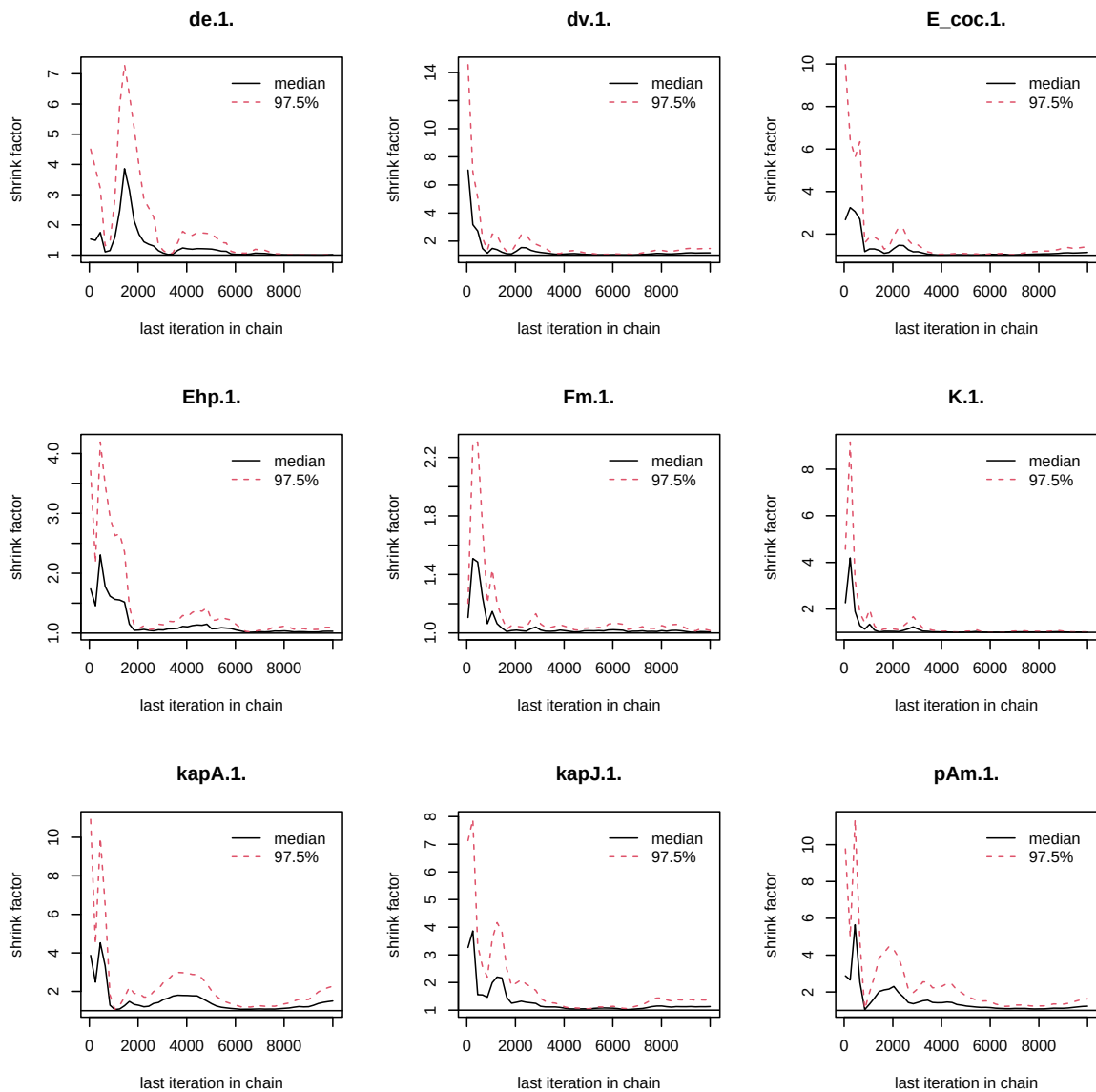


Figure 3: Chains convergence.

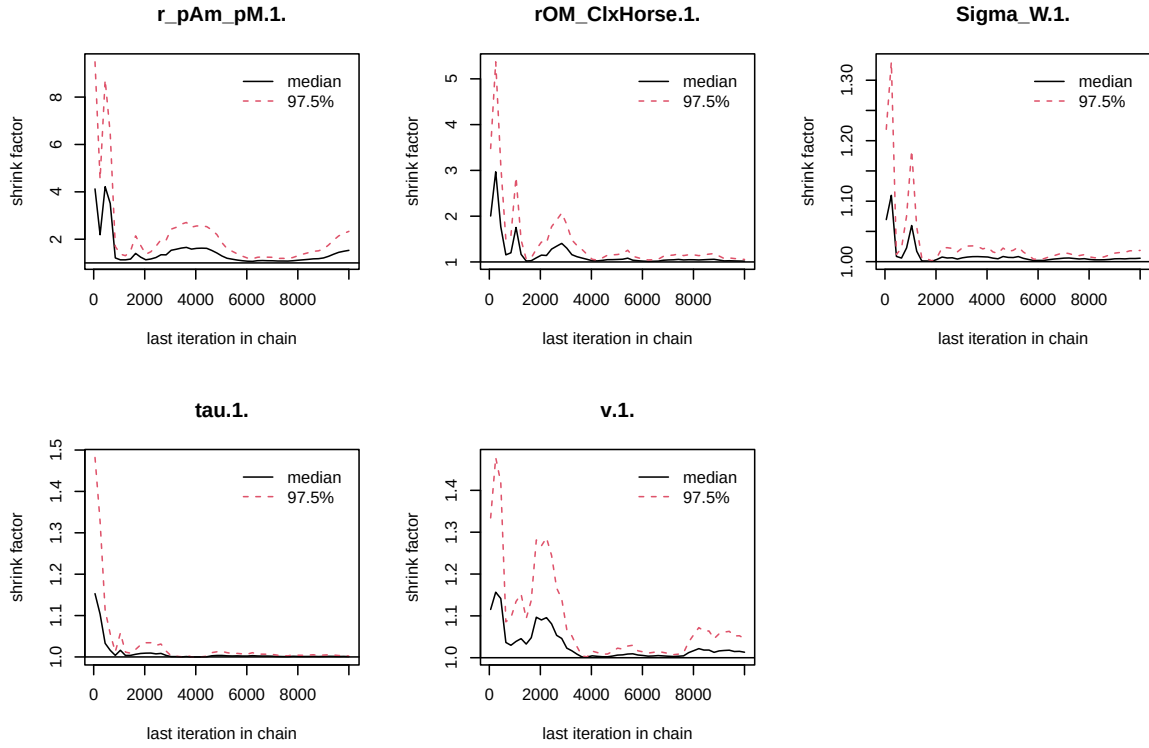


Figure 4: Chains convergence.

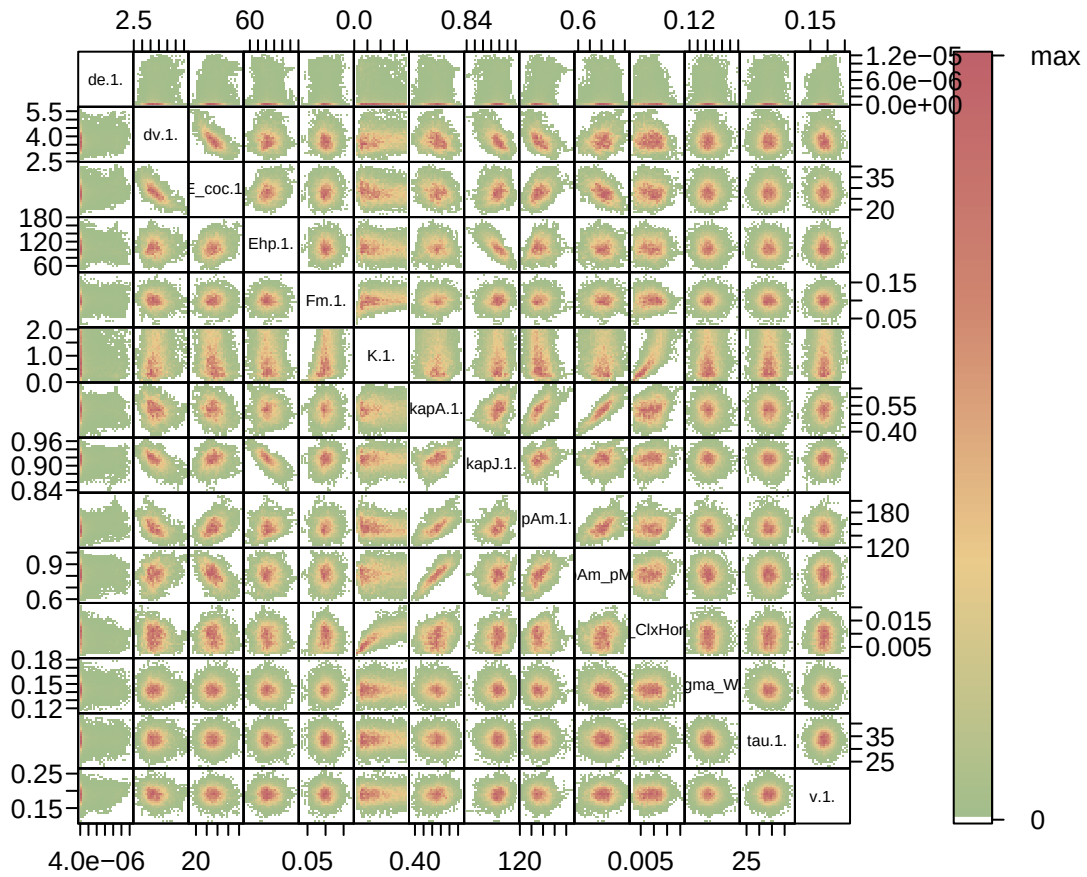


Figure 5: Parameter correlations.

4. Simulations

```
l_Experiments <- c(
  "Bart2019_XP1_1",
  "Bart2019_XP1_2",
  "Bart2019_XP2",
  "Bart2019_XP3",
  "Bart2019_XP4_1",
  "Bart2019_XP4_2",
  "Bart2019_XP5",
```

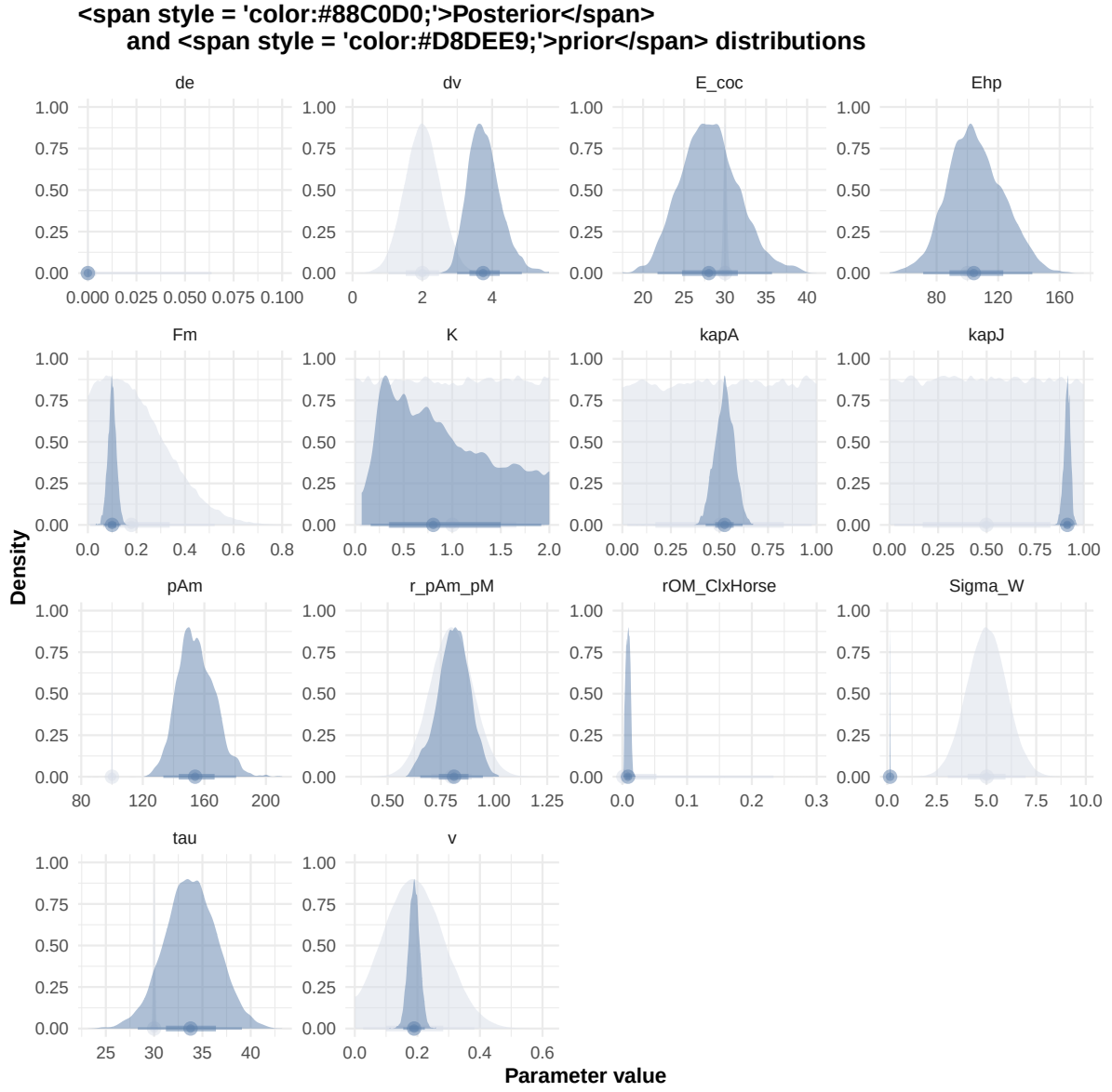


Figure 6: Distributions of the priors and the posteriors.


```
#####
df_No_sim <- df_data |>
  dplyr::select(ID_experiment, No_sim)

df_carac_color <- data.frame(
  carac = unique(l_carac),
  color = c(Nord_aurora[1], Nord_polar[2], Nord_aurora[3], Nord_aurora[2], Nord_aurora[5],
            Nord_aurora[5], Nord_frost[4], Nord_frost[3], Nord_frost[2], Nord_frost[1])
)
v_color_carac <- setNames(df_carac_color$color, df_carac_color$carac)
#####

df_data <- f_import_data_DEB(l_Experiments, Adults_alone) |>
  left_join(df_carc_exp, by="ID_experiment")

# time_limits <- df_data %>%
#   group_by(ID_experiment) %>%
#   summarise(time_cutoff = max(Time, na.rm = TRUE) + 5)

select_times <- seq(0,400, 0.1)

df_sim <- f_MCSim_read_sim(here::here(file_path, "sim.out")) |>
  left_join(df_No_sim, by="No_sim") |>
  left_join(df_carc_exp, by="ID_experiment") |>
  filter(Time %in% select_times) |>
  filter(!((Time > 70) & (ID_experiment == "Gollot2026_dens_D7"))) |>
  mutate(
    Reproduction = case_when(
      Reproduction <= 5e-6 ~ 0,
      .default = Reproduction
    )
  )
)
```

Warning: `read\_table2()` was deprecated in readr 2.0.0.
 i Please use `read\_table()` instead.

Warning in left_join(f_MCSim_read_sim(here::here(file_path, "sim.out")), : Detected an unexpected relationship between variables
 i Row 1 of `x` matches multiple rows in `y`.
 i Row 1 of `y` matches multiple rows in `x`.
 i If a many-to-many relationship is expected, set `relationship = "many-to-many"` to silence this warning.

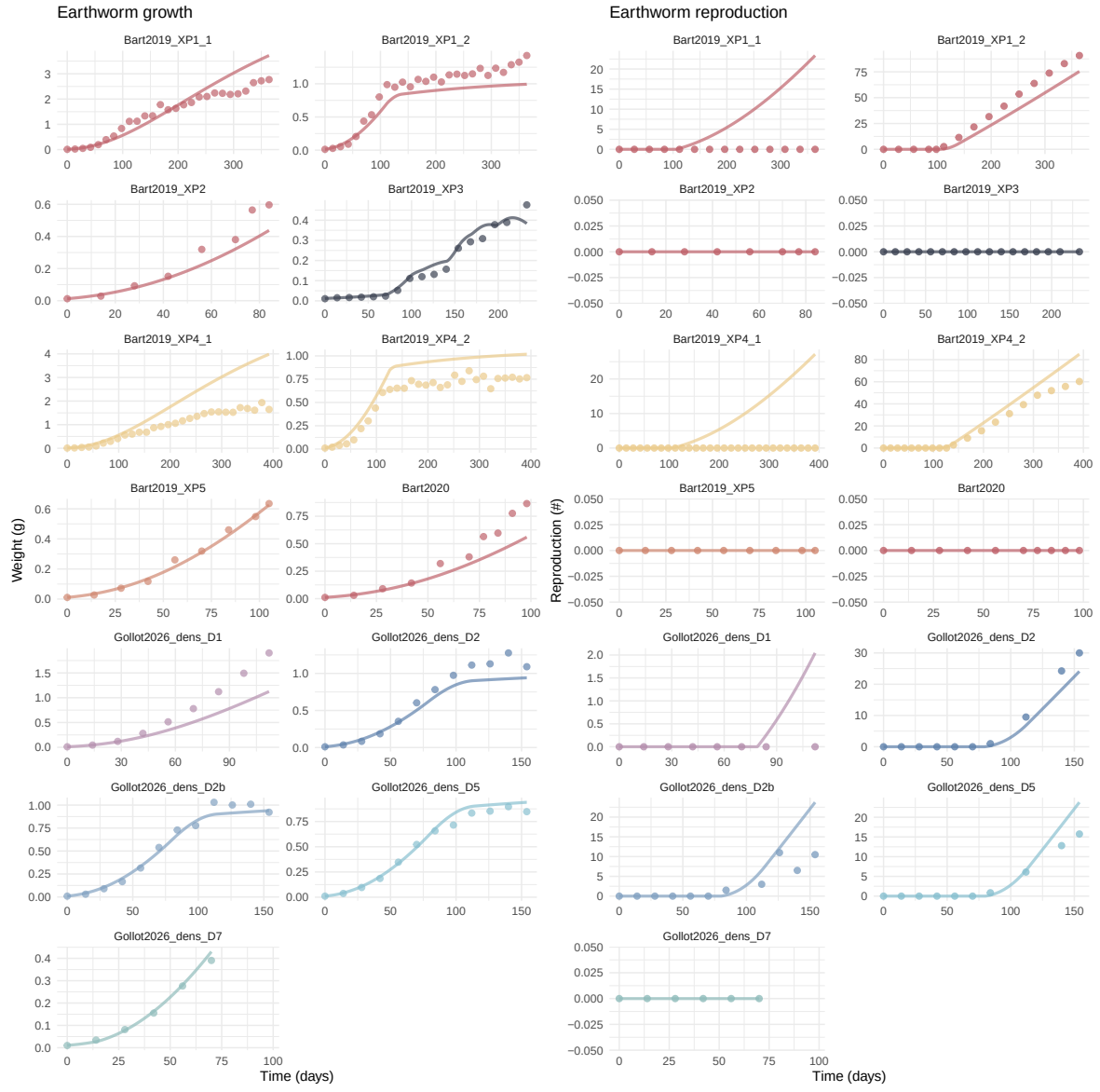
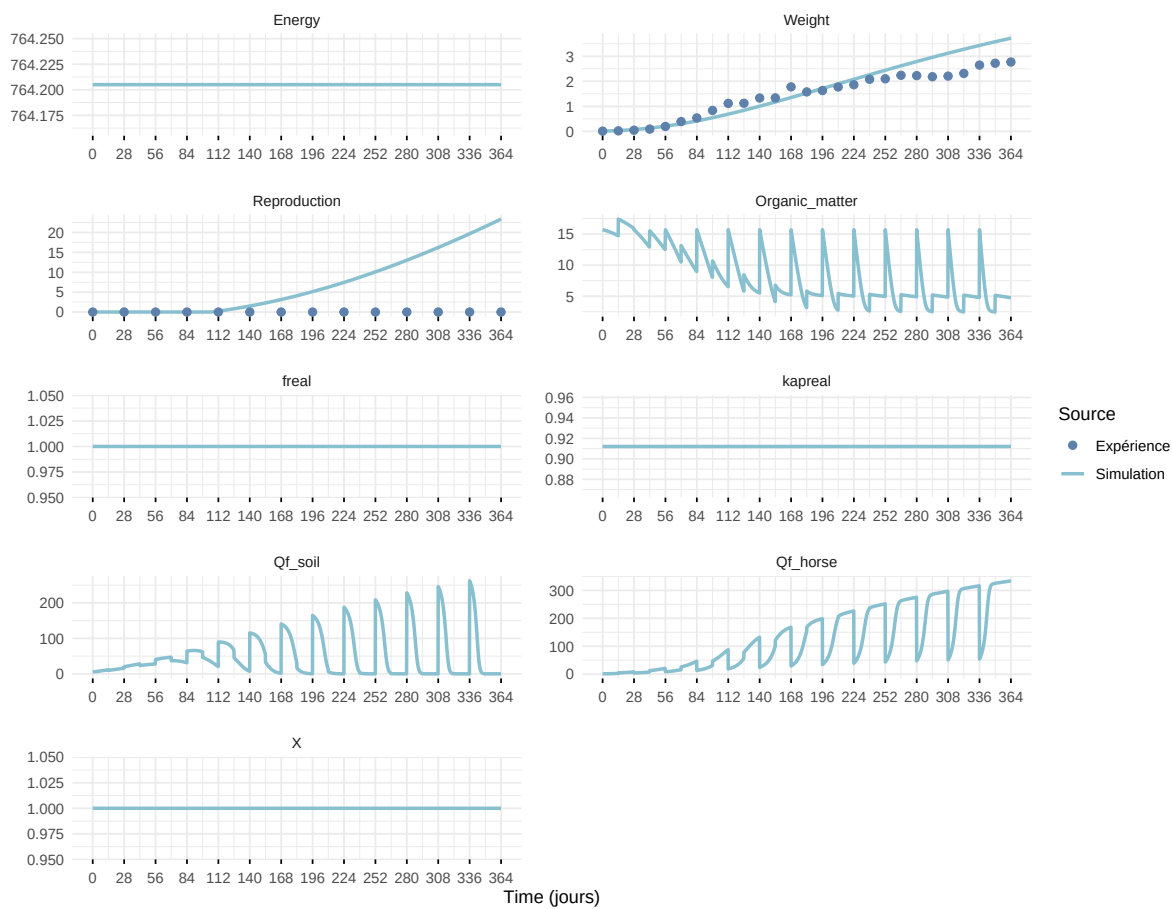
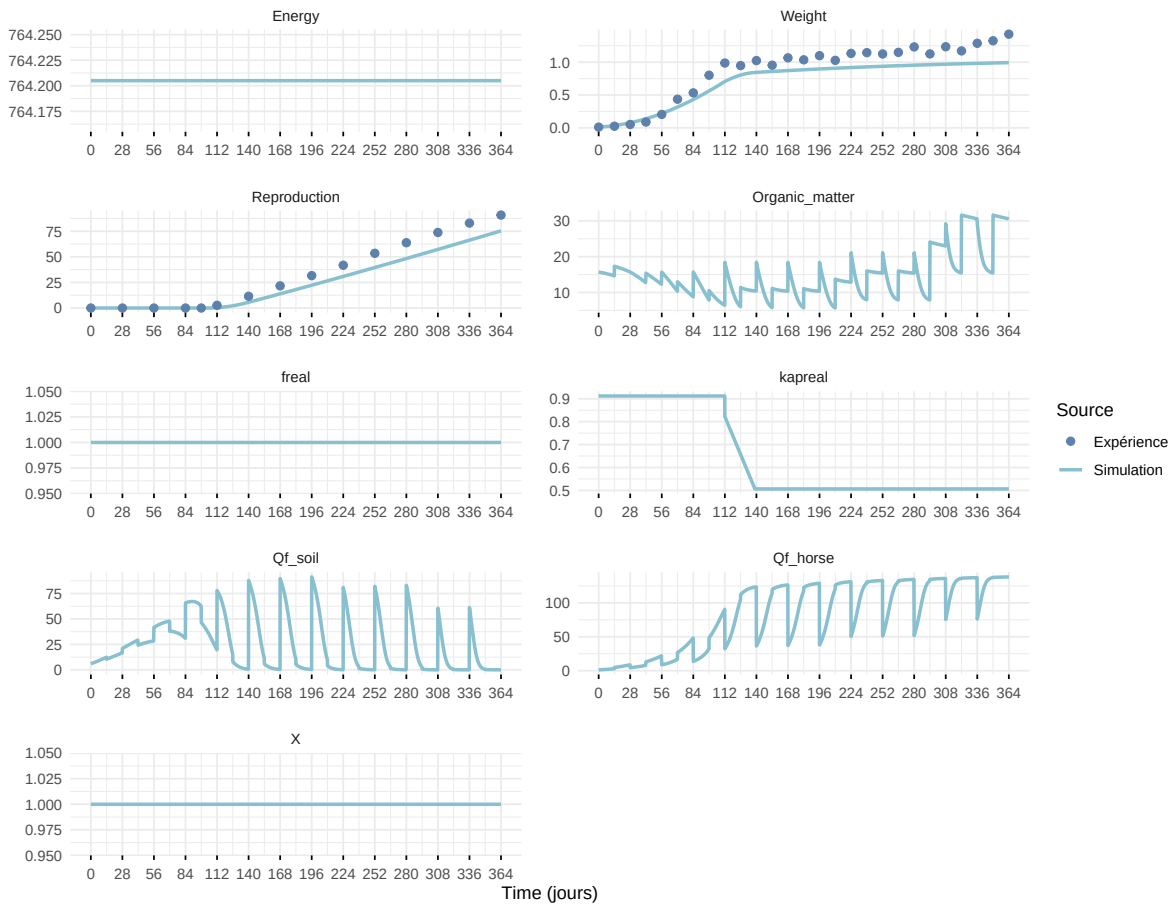


Figure 7: Simulations Weight and Reproduction.

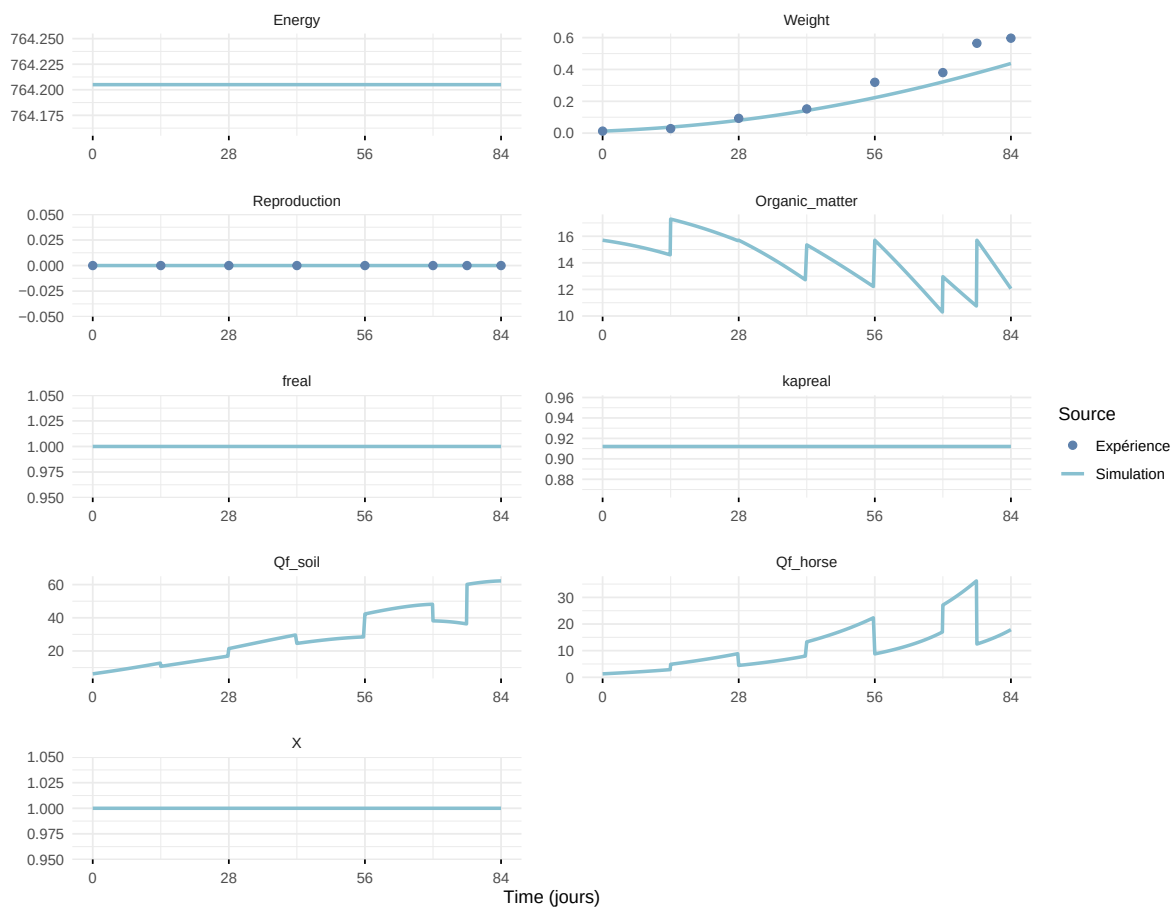
Expérience : Bart2019_XP1_1



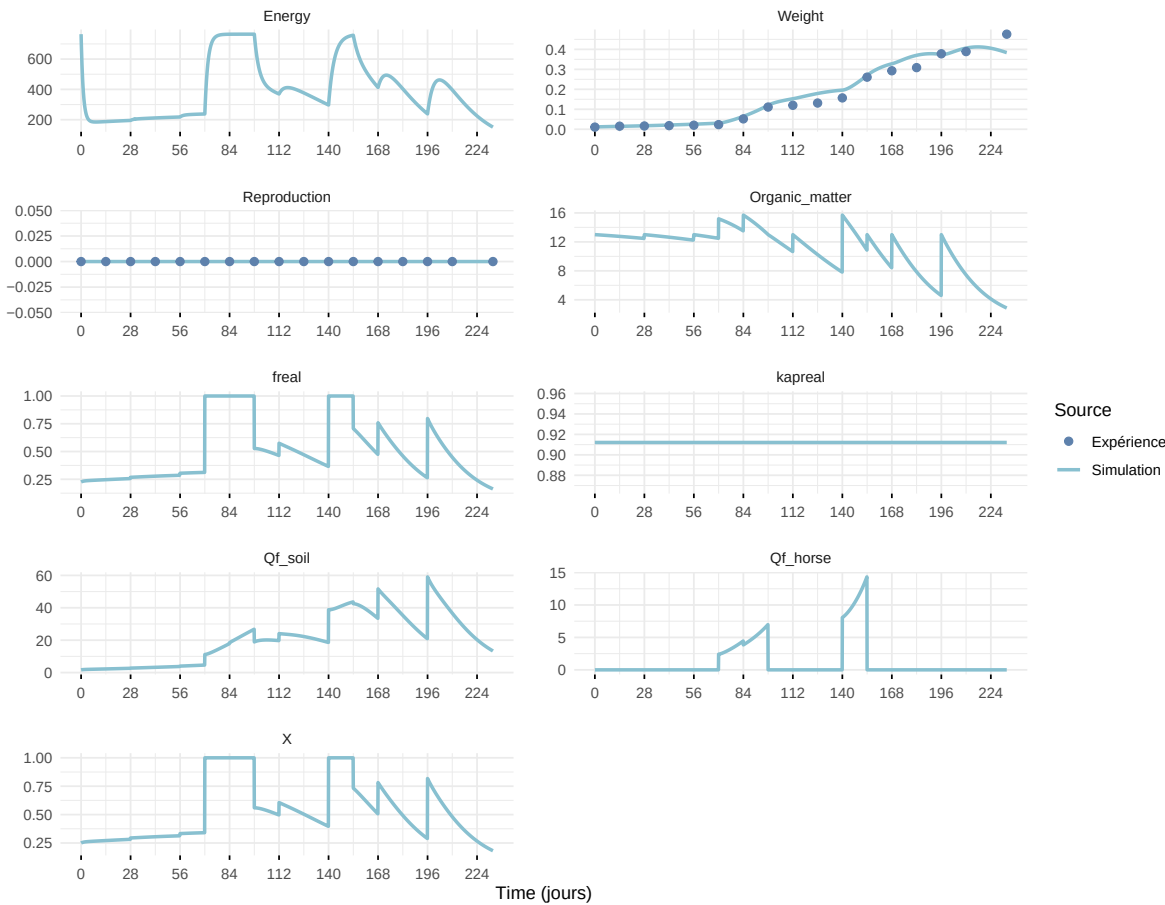
Expérience : Bart2019_XP1_2



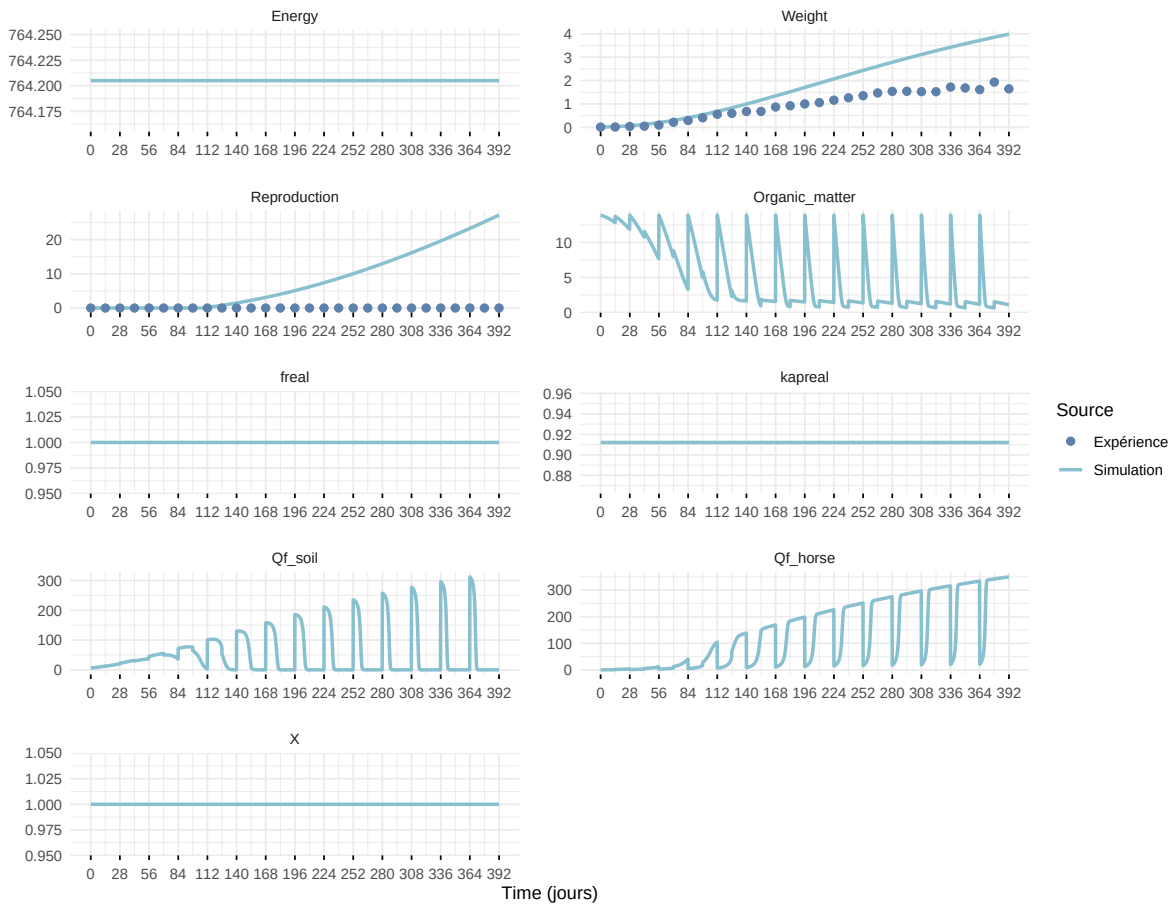
Expérience : Bart2019_XP2



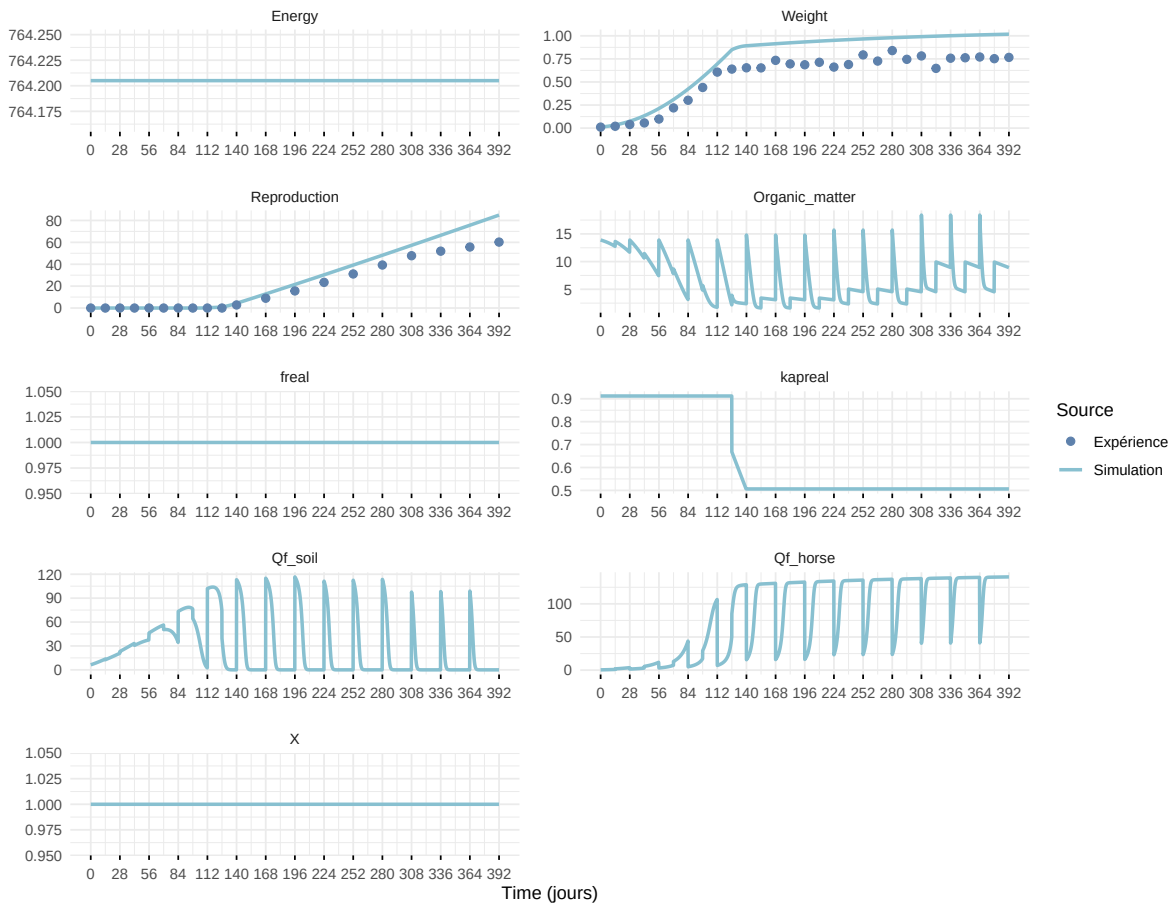
Expérience : Bart2019_XP3



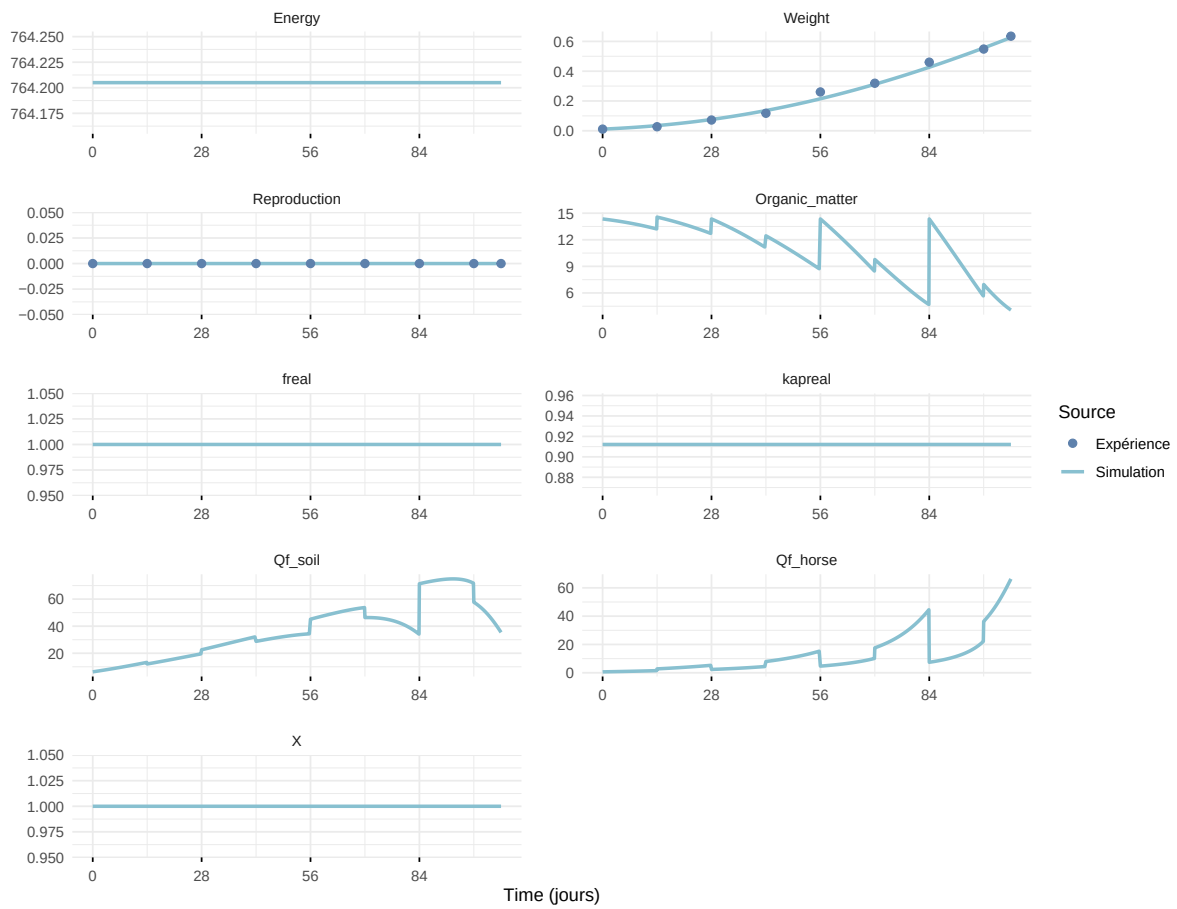
Expérience : Bart2019_XP4_1



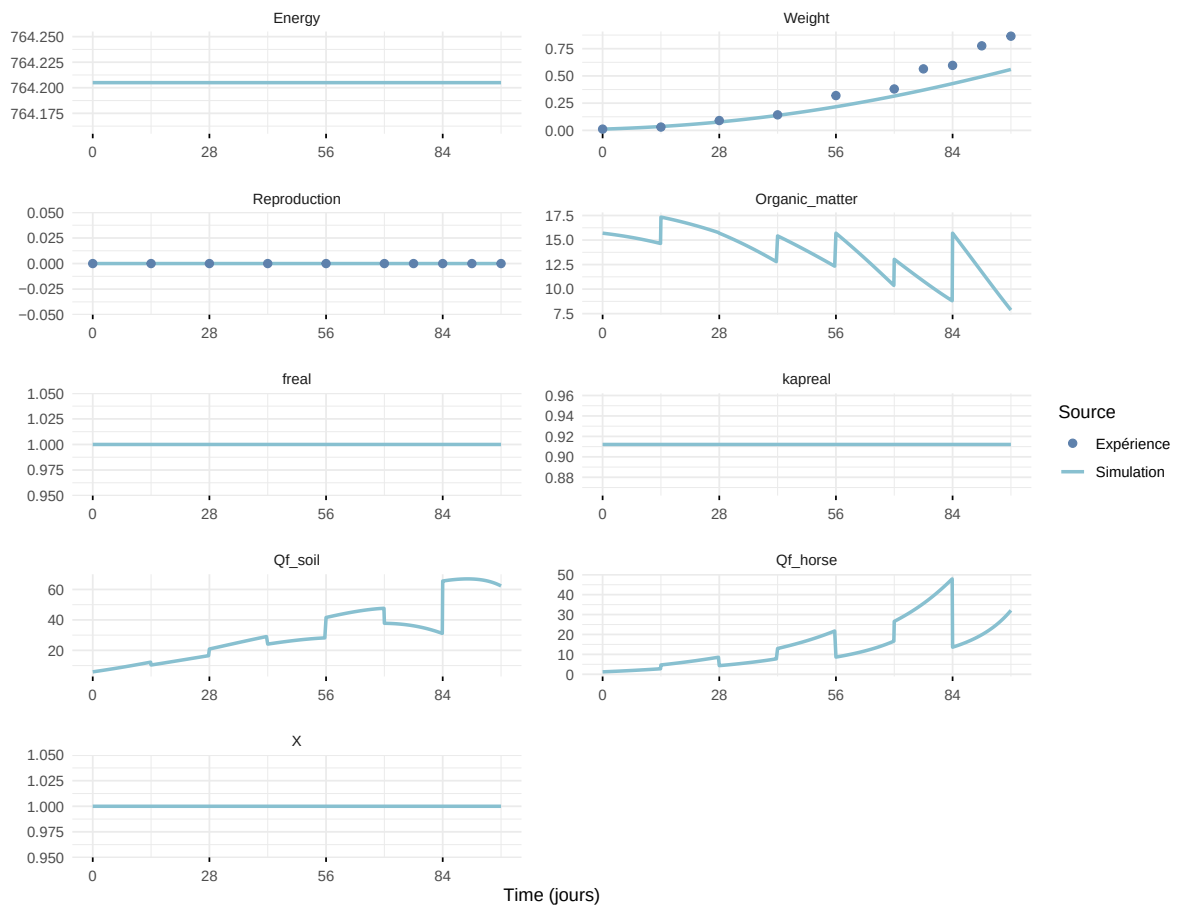
Expérience : Bart2019_XP4_2



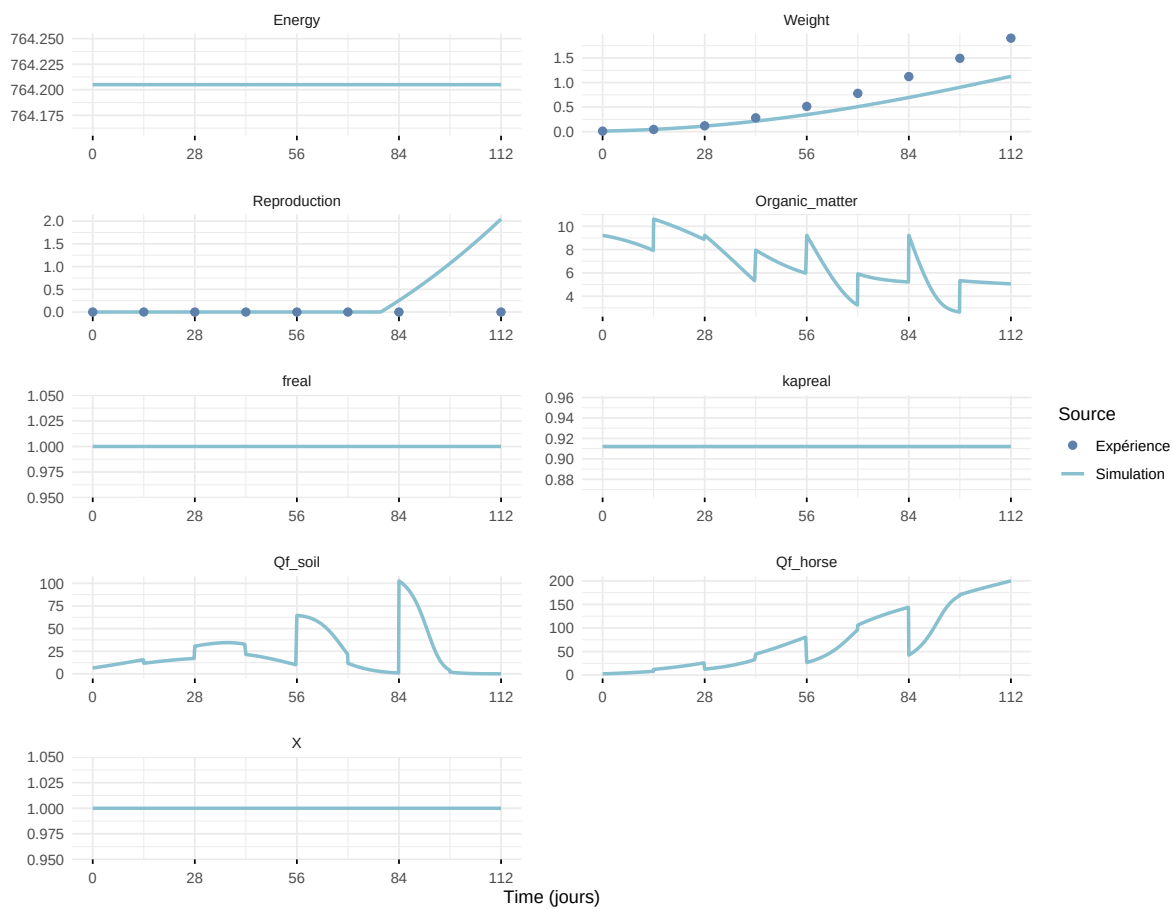
Expérience : Bart2019_XP5



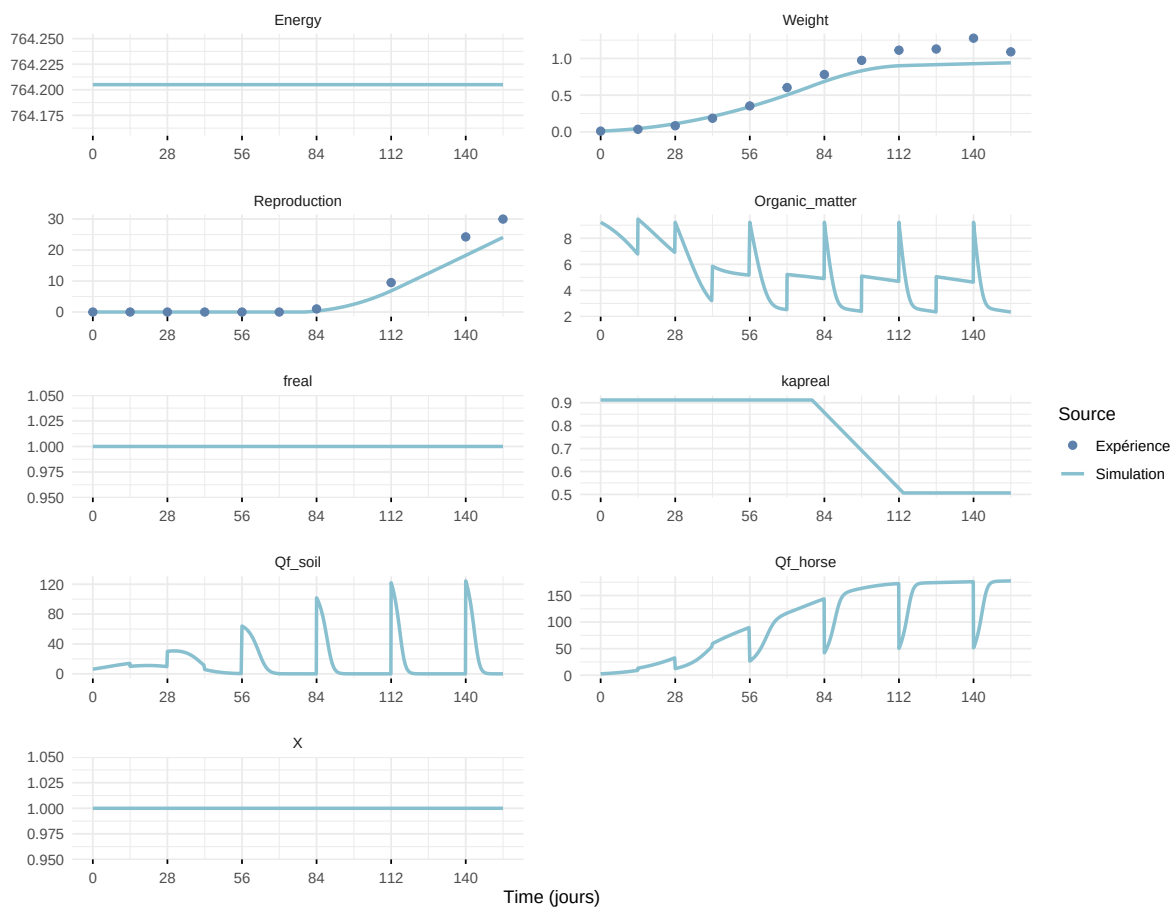
Expérience : Bart2020



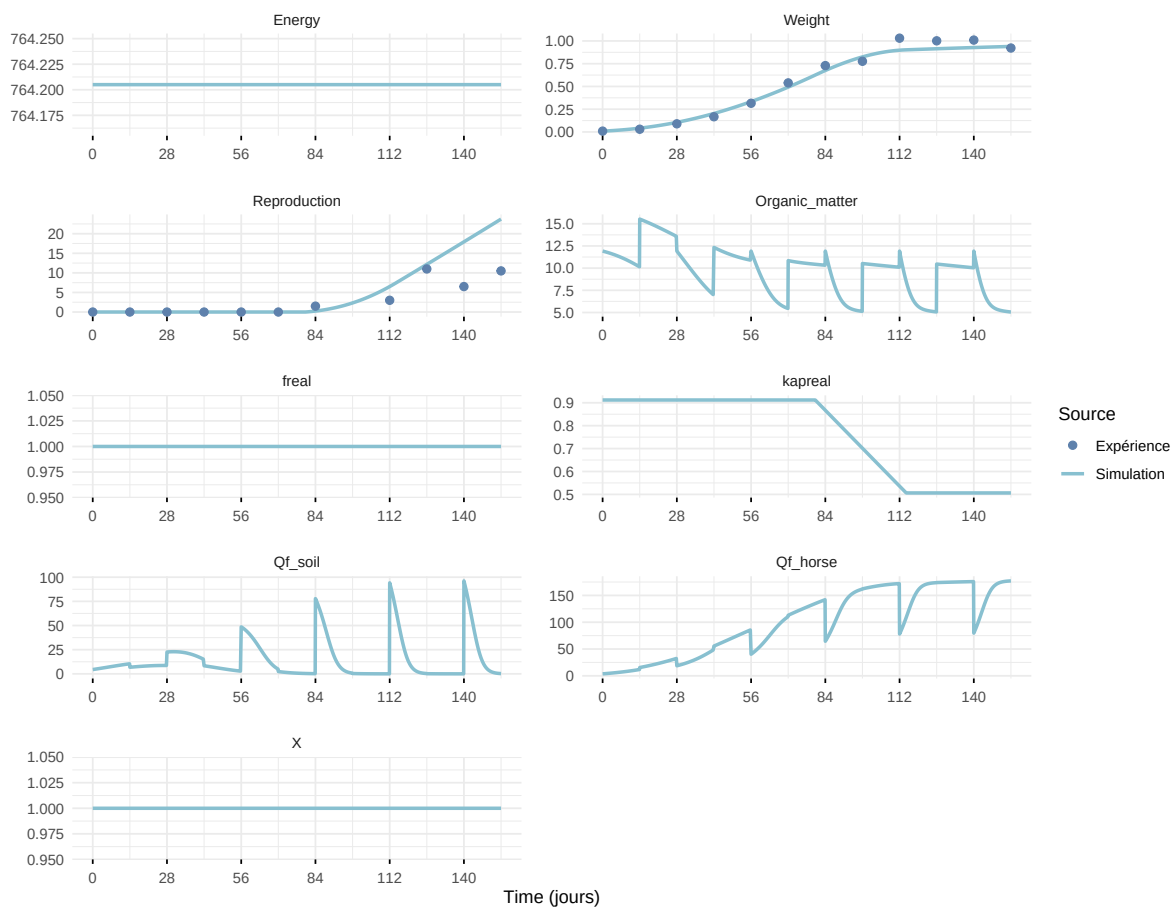
Expérience : Gollot2026_dens_D1



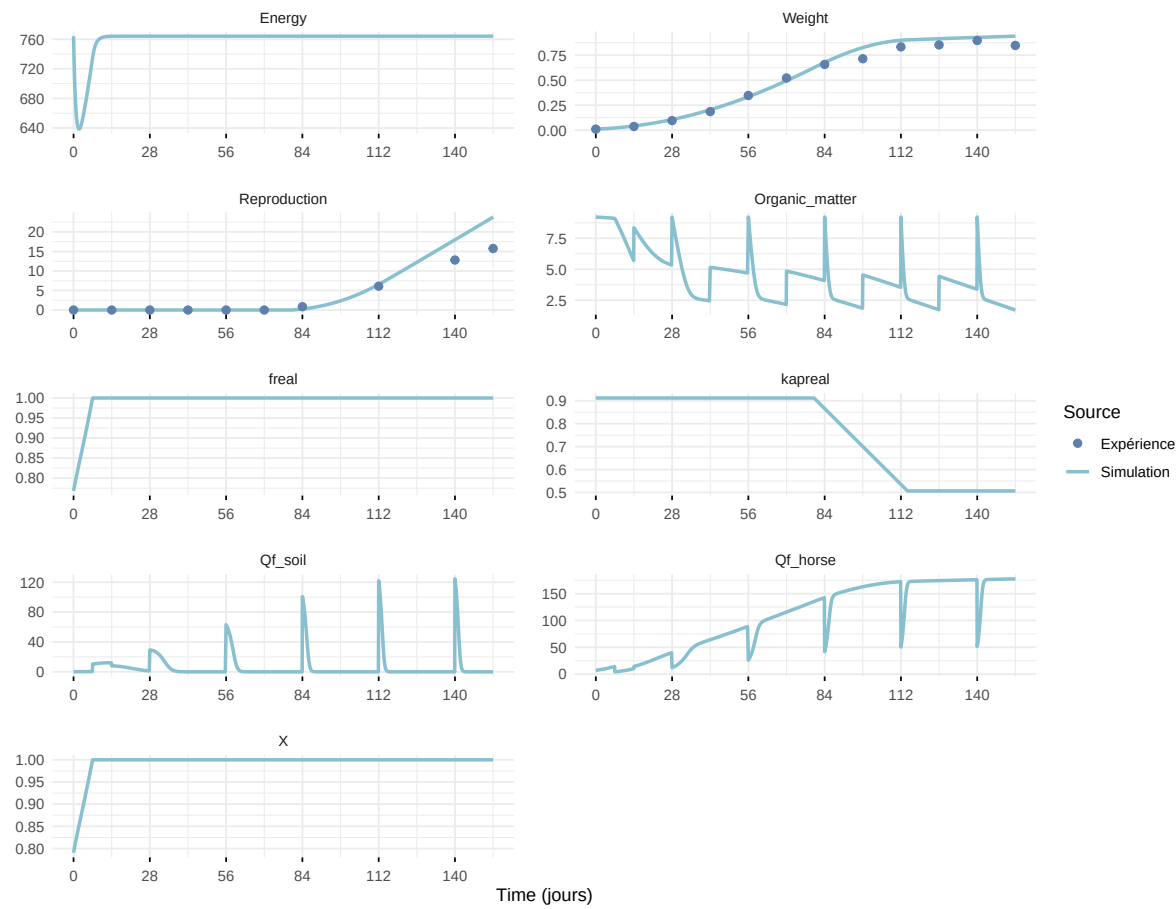
Expérience : Gollot2026_dens_D2



Expérience : Gollot2026_dens_D2b



Expérience : Gollot2026_dens_D5



Expérience : Gollot2026_dens_D7

